

Classification Model

- Classification: Basic Concepts
- Rule-Based Classification
- Decision Tree Induction
- Bayes Classification Methods
- kNN Classifier
- Performance Evaluation
- Ensemble Classification
- Summary



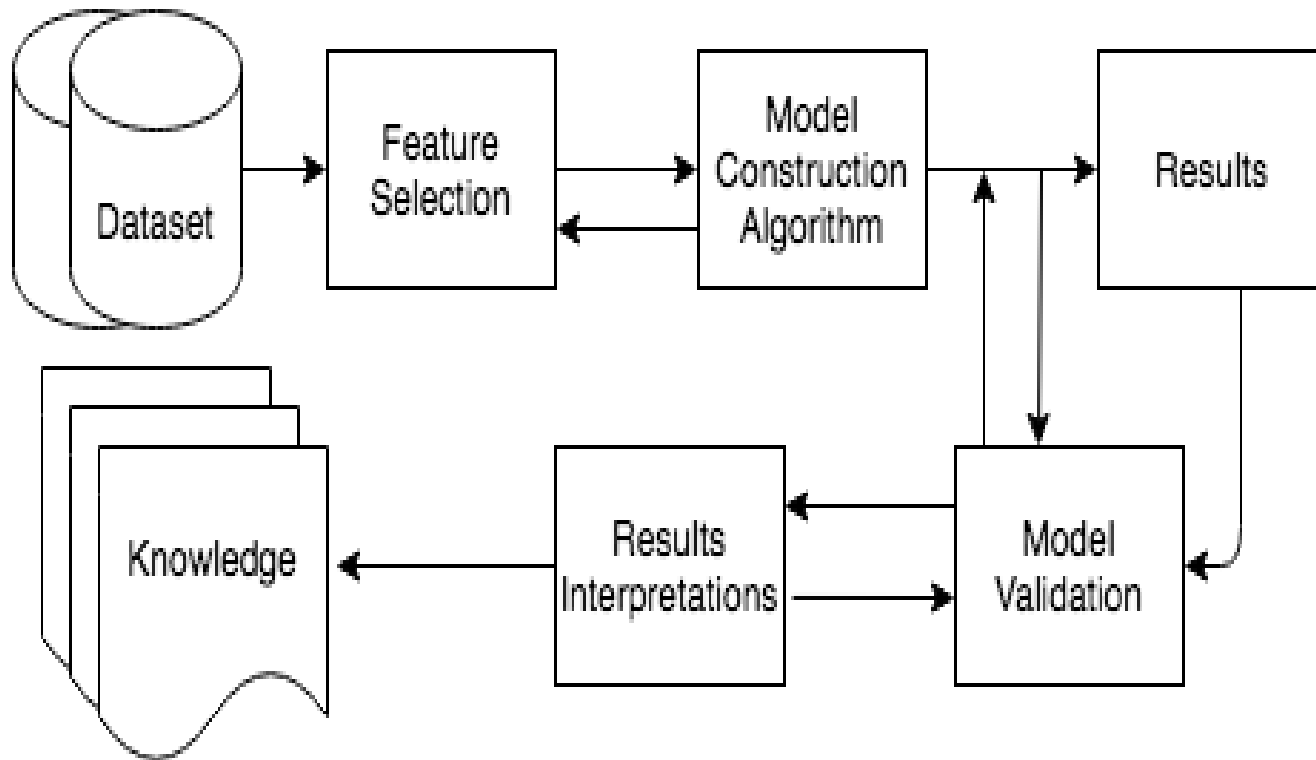
Classification Model

- Supervised learning (classification)
 - Supervision: The training data (observations, measurements, etc.) are accompanied by **labels** indicating the class of the observations
 - New data is classified based on the training set
- Unsupervised learning (clustering)
 - The class labels of training data is unknown
 - Given a set of measurements, observations, etc. with the aim of establishing the existence of classes or clusters in the data

Prediction Problems: Classification vs. Numeric Prediction

- Classification
 - predicts categorical class labels (discrete or nominal)
 - classifies data (constructs a model) based on the training set and the values (**class labels**) in a classifying attribute and uses it in classifying new data
- Numeric Prediction
 - models continuous-valued functions, i.e., predicts unknown or missing values - Regression
- Typical applications
 - Credit/loan approval: if a loan is approved or not
 - Medical diagnosis: if a tumor is cancerous or benign
 - Fraud detection: if a transaction is fraudulent
 - Web page categorization: which category it is

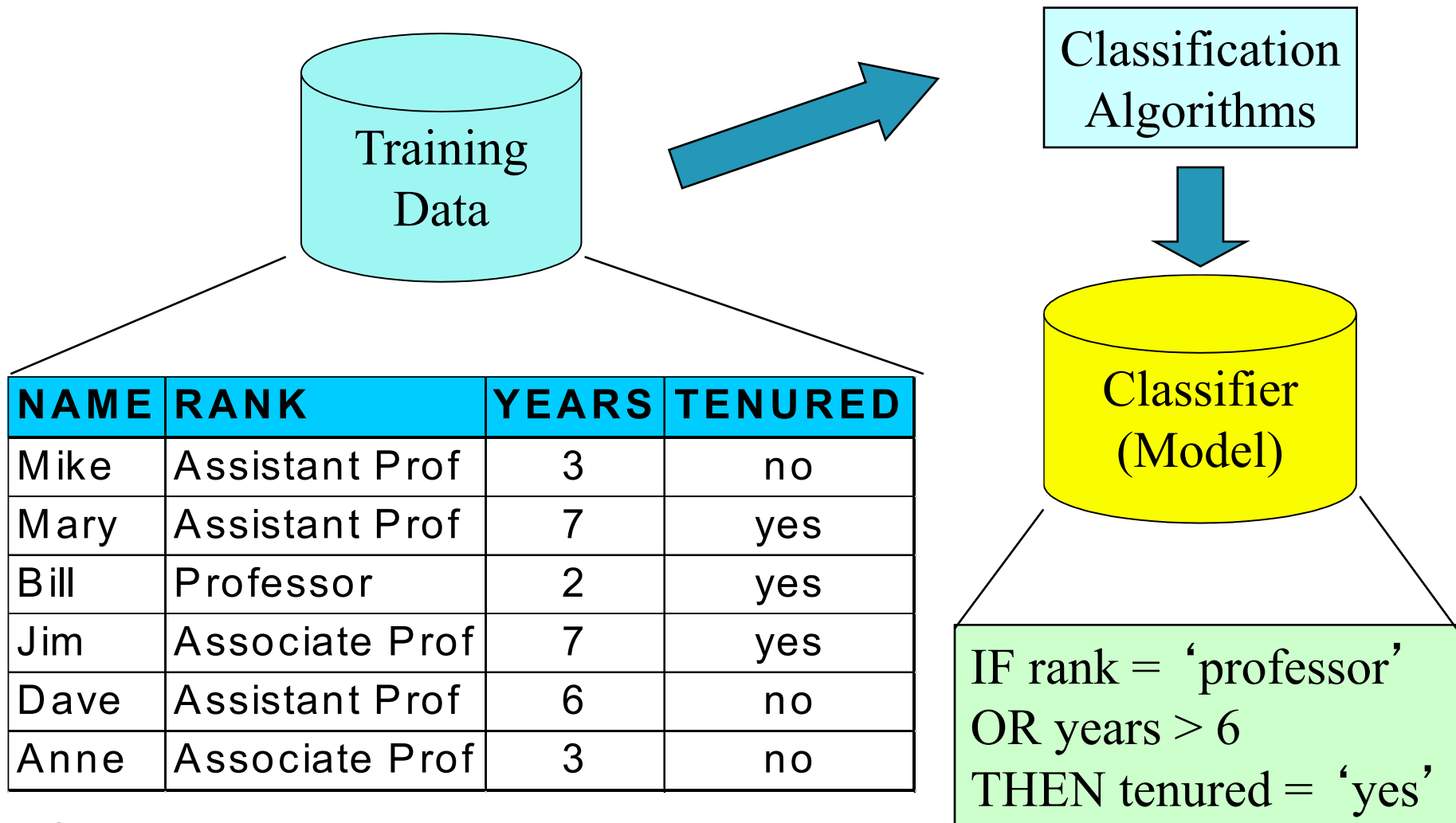
Basic Steps of Decision Making System



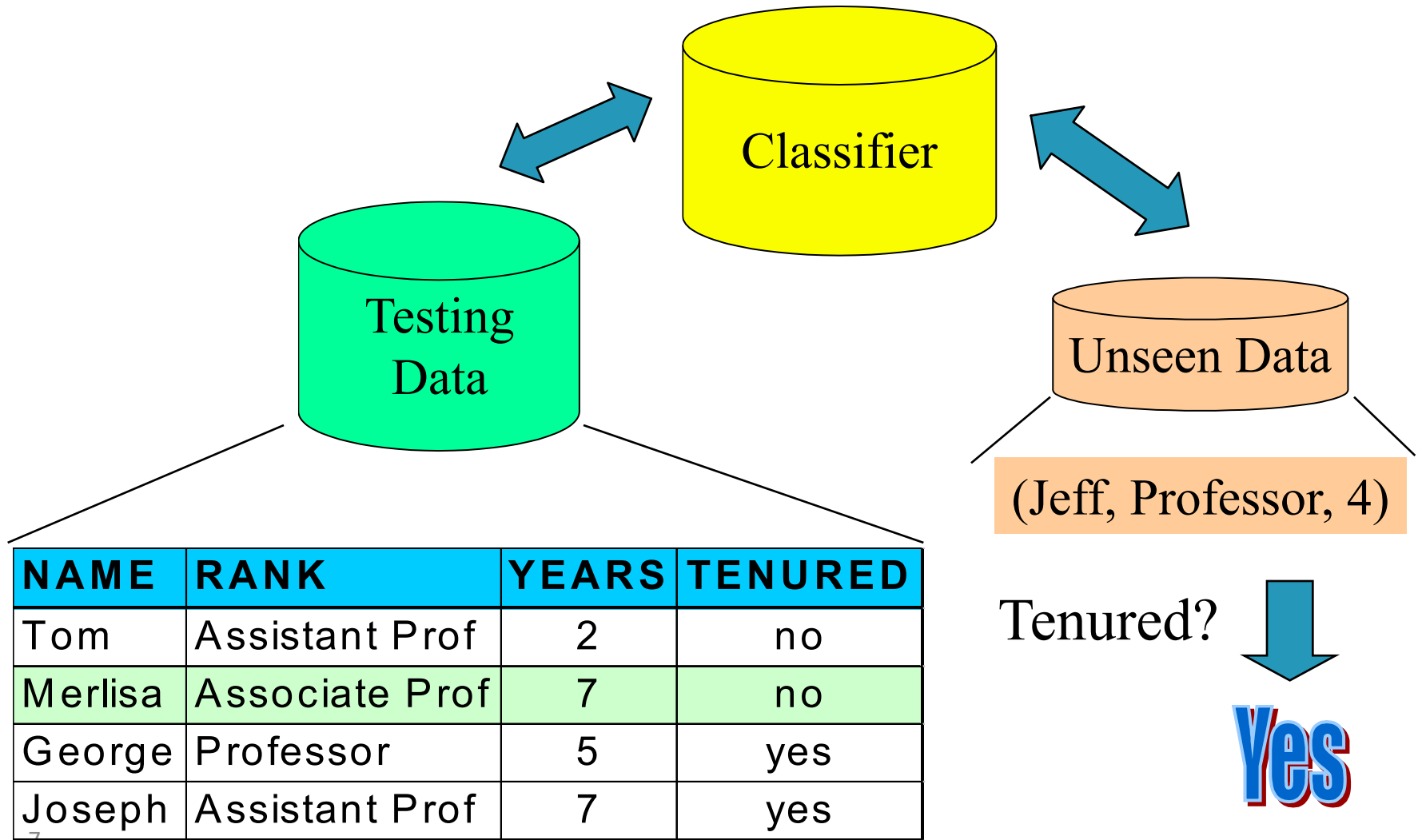
Classification—A Two-Step Process

- **Model construction**: describing a set of predetermined classes
 - Each tuple/sample is assumed to belong to a predefined class, as determined by the **class label attribute**
 - The set of tuples used for model construction is **training set**
 - The model is represented as classification rules, decision trees, or mathematical formulae
- **Model usage**: for classifying future or unknown objects
 - **Estimate accuracy** of the model
 - The known label of test sample is compared with the classified result from the model
 - **Accuracy** rate is the percentage of test set samples that are correctly classified by the model
 - **Test set** is independent of training set (otherwise overfitting)
 - If the accuracy is acceptable, use the model to **classify new data**
- Note: If *the test set* is used to select models, it is called **validation (test) set**

Process (1): Model Construction

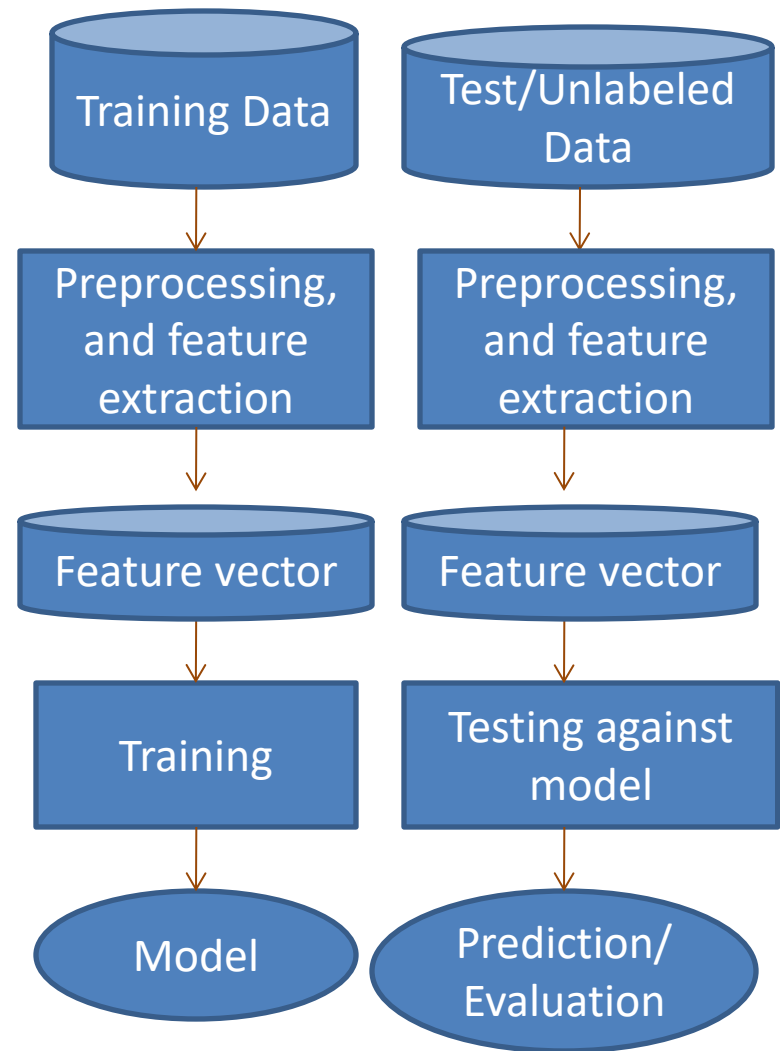


Process (2): Using the Model in Prediction



Application of Classifier in a Data Set

- So the overall classification process goes like this →



Rule based Classifier – Rule Extraction

- What is generally desired is an *intentional* description of the category, a representation of the category based on a set of *rules* that describe the scope of the category.
- The choice of such rules is not unique.
- We will discuss a decision matrix based rule-extraction procedure (Proposed by Ziarko & Shan in 1995).

Rule Extraction

- Let, we wish to find the minimal set of consistent rules called logical implications that characterize our decision system.
- For a set of *condition* attributes, $C = \{A_1, A_2, \dots, A_n\}$ and a decision attribute D , these rules should have the form, $A_1^a A_2^b \dots A_k^p \rightarrow D^d$ or spelled out,

$$(A_1 = a) \wedge (A_2 = b) \wedge \dots \wedge (A_k = p) \rightarrow (D = d)$$

where, $\{a, b, \dots, p\}$ are legitimate values from the domains of their respective attributes.

Rule Extraction

- This is a form typical of association rules, and the number of objects in U which match the condition/antecedent is called the *support* for the rule.
- The method for extracting such rules is to form a *decision matrix* corresponding to each individual value d of decision attribute D .

Decision Matrix

- Informally, the decision matrix for value d of decision attribute D lists all attribute–value pairs that *differ* between objects having $D = d$ and $D \neq d$
- Example – Consider a dataset and say, it is reduced by a feature selection Algorithm, as shown in the right side.

A ₁	A ₂	A ₃	A ₄	D
M	L	3	M	1
M	L	1	H	1
L	L	1	M	1
L	R	3	M	2
M	R	2	M	2
L	R	3	L	3
H	R	3	L	3
H	N	3	L	3

?

Horizontally &
Vertically
Reduced



A ₂	A ₄	D
L	M	1
L	H	1
R	M	2
R	L	3
N	L	3

?

Decision Matrix

A_2	A_4	D
L	M	1
L	H	1
R	M	2
R	L	3
N	L	3

- Decision variable D takes on three different values, namely $\{1, 2, 3\}$. We treat each case separately.

Case-1: $D = 1$

- we divide the objects based on $D = 1$ and $D \neq 1$
- The objects having $D = 1$ are $\{O_1, O_2\}$ while the objects having $D \neq 1$ are $\{O_3, O_4, O_5\}$.

Decision Matrix for $D=1$	O_3	O_4	O_5
O_1	A_2^L	$A_2^L A_4^M$	$A_2^L A_4^M$
O_2	$A_2^L A_4^H$	$A_2^L A_4^H$	$A_2^L A_4^H$

- The decision matrix for $D = 1$ lists all the differences between the objects having $D = 1$ and those having $D \neq 1$.
- Thus, the decision matrix lists all the differences between $\{O_1, O_2\}$ and $\{O_3, O_4, O_5\}$.

Decision Matrix

- Let, $A_2^L = x, A_4^M = y, A_4^H = z$
- Thus, for this table we have following two Boolean expressions:

Decision Matrix for D=1	O ₃	O ₄	O ₅
O ₁	A_2^L	$A_2^L A_4^M$	$A_2^L A_4^M$
O ₂	$A_2^L A_4^H$	$A_2^L A_4^H$	$A_2^L A_4^H$

- $x(x+y)(x+y) = x(x+y) = x = A_2^L$
- $(x+z)(x+z)(x+z) = x+z = A_2^L + A_4^H$

- Thus we have following two rules:

Rule 1: If $A_2 = L$ Then $D = 1$

Rule 2: If $A_2 = L$ OR $A_4 = H$ Then $D=1$

Decision Matrix

Case-2: $D = 2$

A_2	A_4	D
L	M	1
L	H	1
R	M	2
R	L	3
N	L	3

2

- we divide the objects based on $D = 2$ and $D \neq 2$
- The object having $D = 2$ is $\{O_3\}$ while the objects having $D \neq 2$ are $\{O_1, O_2, O_4, O_5\}$.
- The decision matrix for $D = 2$ lists all the differences between the objects having $D = 2$ and those having $D \neq 2$.
- Thus, the decision matrix lists all the differences between $\{O_3\}$ and $\{O_1, O_2, O_4, O_5\}$.

Decision Matrix for $D=2$	O_1	O_2	O_4	O_5
O_3	A_2^R	$A_2^R A_4^M$	A_4^M	$A_2^R A_4^M$

Decision Matrix

- Let, $A_2^R = x, A_4^M = y$
- Thus, for this table we have following Boolean expression:

Decision Matrix for D=2	O ₁	O ₂	O ₄	O ₅
O ₃	A_2^R	$A_2^R A_4^M$	A_4^M	$A_2^R A_4^M$

1. $x(x+y)y(x+y) = xy(x+y) = xy + xy = xy = A_2^R A_4^M$

- Thus we have the following rule:

Rule 3: If $A_2 = R$ AND $A_4 = M$ Then $D = 2$

Decision Matrix

Case-3: $D = 3$

A_2	A_4	D
L	M	1
L	H	1
R	M	2
R	L	3
N	L	3

?

- we divide the objects based on $D = 3$ and $D \neq 3$
- The objects having $D = 3$ are $\{O_4, O_5\}$ while the objects having $D \neq 3$ are $\{O_1, O_2, O_3\}$.
- The decision matrix for $D = 3$ lists all the differences between the objects having $D = 3$ and those having $D \neq 3$.

Decision Matrix for $D=3$	O_1	O_2	O_3
O_4	$A_2^R A_4^L$	$A_2^R A_4^L$	A_4^L
O_5	$A_2^N A_4^L$	$A_2^N A_4^L$	$A_2^N A_4^L$

- Thus, the decision matrix lists all the differences between $\{O_4, O_5\}$ and $\{O_1, O_2, O_3\}$.

Decision Matrix

- Let, $A_2^R = x, A_4^L = y, A_2^N = z$
- Thus, for this table we have following two Boolean expressions:

Decision Matrix for D=3	O ₁	O ₂	O ₃
O ₄	$A_2^R A_4^L$	$A_2^R A_4^L$	A_4^L
O ₅	$A_2^N A_4^L$	$A_2^N A_4^L$	$A_2^N A_4^L$

- $(x+y)(x+y)y = y(x+y) = y = A_4^L$
- $(z+y)(z+y)(z+y) = z+y = A_2^N + A_4^L$

- Thus we have following two rules:

Rule 4: If $A_4 = L$ Then $D = 3$

Rule 5: If $A_2 = N$ OR $A_4 = L$ Then $D=3$

Decision Rules

- Thus we have following rules for this data set:

Rule 1: $(A_2 = L) \Rightarrow (D = 1)$

Rule 2: $(A_2 = L) \text{ OR } (A_4 = H) \Rightarrow (D=1)$

Rule 3: $(A_2 = R) \text{ AND } (A_4 = M) \Rightarrow (D = 2)$

Rule 4: $(A_4 = L) \Rightarrow (D = 3)$

Rule 5: $(A_2 = N) \text{ OR } (A_4 = L) \Rightarrow (D=3)$

Here, Training Accuracy = 100%

A_2	A_4	D
L	M	1
L	H	1
R	M	2
R	L	3
N	L	3

?

A_1	A_2	A_3	A_4	D
M	L	3	M	1
M	L	1	H	1
L	L	1	M	1
L	R	3	M	2
M	R	2	M	2
L	R	3	L	3
H	R	3	L	3
H	N	3	L	3

?

Data reduction by another feature selection algorithm and Rule based classifier construction

- Informally, the decision matrix for value d of decision attribute D lists all attribute–value pairs that *differ* between objects having $D = d$ and $D \neq d$
- Example – Consider the dataset and reduced dataset

A ₁	A ₂	A ₃	A ₄	D
M	L	3	M	1
M	L	1	H	1
L	L	1	M	1
L	R	3	M	2
M	R	2	M	2
L	R	3	L	3
H	R	3	L	3
H	N	3	L	3

?

Vertically Reduced

A ₁	A ₂	A ₄	D
M	L	M	1
M	L	H	1
L	L	M	1
L	R	M	2
M	R	M	2
L	R	L	3
H	R	L	3
H	N	L	3

?

Decision Matrix

Decision variable D takes on three different values, namely $\{1, 2, 3\}$. We treat each case separately.

A_1	A_2	A_4	D
M	L	M	1
M	L	H	1
L	L	M	1
L	R	M	2
M	R	M	2
L	R	L	3
H	R	L	3
H	N	L	3

2

Case-1: $D = 1$

- We divide the objects based on $D = 1$ and $D \neq 1$
- The objects having $D = 1$ are $\{O_1, O_2, O_3\}$ while the objects having $D \neq 1$ are $\{O_4, O_5, O_6, O_7, O_8\}$.

Decision Matrix for $D=1$	O_4	O_5	O_6	O_7	O_8
O_1	$A_1^M A_2^L$	A_2^L	$A_1^M A_2^L A_4^M$	$A_1^M A_2^L A_4^M$	$A_1^M A_2^L A_4^M$
O_2	$A_1^M A_2^L A_4^H$	$A_2^L A_4^H$	$A_1^M A_2^L A_4^H$	$A_1^M A_2^L A_4^H$	$A_1^M A_2^L A_4^H$
O_3	A_2^L	$A_1^L A_2^L$	$A_2^L A_4^M$	$A_1^L A_2^L A_4^M$	

- The decision matrix for $D = 1$ lists all the differences between the objects having $D = 1$ and those having $D \neq 1$.
- Thus, the decision matrix lists all the differences between $\{O_1, O_2, O_3\}$ and $\{O_4, O_5, O_6, O_7, O_8\}$.

Decision Matrix

Decision Matrix for D=1	O ₄	O ₅	O ₆	O ₇	O ₈
O ₁	$A_1^M A_2^L$	A_2^L	$A_1^M A_2^L A_4^M$	$A_1^M A_2^L A_4^M$	$A_1^M A_2^L A_4^M$
O ₂	$A_1^M A_2^L A_4^H$	$A_2^L A_4^H$	$A_1^M A_2^L A_4^H$	$A_1^M A_2^L A_4^H$	$A_1^M A_2^L A_4^H$
O ₃	A_2^L	$A_1^L A_2^L$	$A_2^L A_4^M$	$A_1^L A_2^L A_4^M$	$A_1^L A_2^L A_4^M$

- Let, $A_1^M = u, A_2^L = v, A_4^M = w, A_4^H = x, A_1^L = y$
- Thus, for this table we have following three Boolean expressions:

- $(u+v)v(u+v+w)(u+v+w)(u+v+w)=v(u+v)(u+v+w)=v = A_2^L$
- $(u+v+x)(v+x)(u+v+x)(u+v+x)(u+v+x)=v+x = A_2^L + A_4^H$
- $v(v+y)(v+w)(y+v+w)(y+v+w)=v(v+y)(v+y+w)=v=A_2^L$

- Thus we have following three rules:

Rule 1: $(A_2 = L) \Rightarrow (D = 1)$

Rule 2: $(A_2 = L) \text{ OR } (A_4 = H) \Rightarrow D=1$

Rule 3: $(A_2 = L) \Rightarrow (D = 1)$ [redundant rule same to Rule 1]

Decision Matrix

Case-2: $D = 2$

A_1	A_2	A_4	D
M	L	M	1
M	L	H	1
L	L	M	1
L	R	M	2
M	R	M	2
L	R	L	3
H	R	L	3
H	N	L	3

- we divide the objects based on $D = 2$ and $D \neq 2$
- The object having $D = 2$ is $\{O_4, O_5\}$ while the objects having $D \neq 2$ are $\{O_1, O_2, O_3, O_6, O_7, O_8\}$.
- The decision matrix for $D = 2$ lists all the differences between the objects having $D = 2$ and those having $D \neq 2$.
- Thus, the decision matrix lists all the differences between $\{O_4, O_5\}$ and $\{O_1, O_2, O_3, O_6, O_7, O_8\}$.

Decision Matrix for $D=2$	O_1	O_2	O_3	O_6	O_7	O_8
O_4	$A_1^L A_2^R$	$A_1^L A_2^R A_4^M$	A_2^R	A_4^M	$A_1^L A_4^M$	$A_1^L A_2^R A_4^M$
O_5	A_2^R	$A_2^R A_4^M$	$A_1^M A_2^R$	$A_1^M A_4^M$	$A_1^M A_4^M$	$A_1^M A_2^R A_4^M$

Decision Matrix

- Let, $A_1^L = u, A_2^R = v, A_4^M = w, A_1^M = x$
- Thus, for this table we have following Boolean expressions:

Decision Matrix for D=2	O ₁	O ₂	O ₃	O ₆	O ₇	O ₈
O ₄	$A_1^L A_2^R$	$A_1^L A_2^R A_4^M$	A_2^R	A_4^M	$A_1^L A_4^M$	$A_1^L A_2^R A_4^M$
O ₅	A_2^R	$A_2^R A_4^M$	$A_1^M A_2^R$	$A_1^M A_4^M$	$A_1^M A_4^M$	$A_1^M A_2^R A_4^M$

- $(u+v)(u+v+w)vw(u+w)(u+v+w) = vw$
- $v(v+w)(x+v)(x+w)(x+w)(x+v+w) = v(x+w) = vx + vw$

- Thus we have the following rule:

Rule 4: $(A_2 = R) \text{ AND } (A_4 = M) \Rightarrow (D=2)$

Rule 5: $((A_2 = R) \text{ AND } (A_1 = M)) \text{ OR } (A_2 = R) \text{ AND } (A_4 = M) \Rightarrow (D=2)$

Decision Matrix

Case-3: $D = 3$

A_1	A_2	A_4	D
M	L	M	1
M	L	H	1
L	L	M	1
L	R	M	2
M	R	M	2
L	R	L	3
H	R	L	3
H	N	L	3

- we divide the objects based on $D = 3$ and $D \neq 3$
- The objects having $D = 3$ are $\{O_6, O_7, O_8\}$ while the objects having $D \neq 3$ are $\{O_1, O_2, O_3, O_4, O_5\}$.
- The decision matrix for $D = 3$ lists all the differences between the objects having $D = 3$ and those having $D \neq 3$.

Decision Matrix for $D=3$	O_1	O_2	O_3	O_4	O_5
O_6	$A_1^L A_2^R A_4^L$	$A_1^L A_2^R A_4^L$	$A_2^R A_4^L$	A_4^L	$A_1^L A_4^L$
O_7	$A_1^H A_2^R A_4^L$	$A_1^H A_2^R A_4^L$	$A_1^H A_2^R A_4^L$	$A_1^H A_4^L$	$A_1^H A_4^L$
O_8	$A_1^H A_2^N A_4^L$	$A_1^H A_2^N A_4^L$	$A_1^H A_2^N A_4^L$	$A_1^H A_2^N A_4^L$	$A_1^H A_2^N A_4^L$

Thus, the decision matrix lists all the differences between $\{O_6, O_7, O_8\}$ and $\{O_1, O_2, O_3, O_4, O_5\}$.

Decision Matrix

Decision Matrix for D=3	O ₁	O ₂	O ₃	O ₄	O ₅
O ₆	$A_1^L A_2^R A_4^L$	$A_1^L A_2^R A_4^L$	$A_2^R A_4^L$	A_4^L	$A_1^L A_4^L$
O ₇	$A_1^H A_2^R A_4^L$	$A_1^H A_2^R A_4^L$	$A_1^H A_2^R A_4^L$	$A_1^H A_4^L$	$A_1^H A_4^L$
O ₈	$A_1^H A_2^N A_4^L$	$A_1^H A_2^N A_4^L$	$A_1^H A_2^N A_4^L$	$A_1^H A_2^N A_4^L$	$A_1^H A_2^N A_4^L$

- Let, $A_1^L = u, A_2^R = v, A_4^L = w, A_1^H = x, A_2^N = y$
- Thus, we have following Boolean expressions:

- $(u+v+w)(u+v+w)(v+w)w(u+w) = w$
- $(x+v+w)(x+v+w)(x+v+w)(x+w)(x+w) = (x+w)$
- $(x+y+w)(x+y+w)(x+y+w)(x+y+w)(x+y+w) = (x+y+w)$

- Thus we have following rules:

Rule 7: $(A_4 = L) \Rightarrow (D = 3)$

Rule 8: $(A_1 = H) \text{ OR } (A_4 = L) \Rightarrow (D=3)$

Rule 9: $(A_1 = H) \text{ OR } (A_2 = N) \text{ OR } (A_4 = L) \Rightarrow (D=3)$

Decision Rules

- Thus we have following rules for this data set:

$$(A_2 = L) \Rightarrow (D = 1)$$

$$(A_2 = L) \text{ OR } (A_4 = H) \Rightarrow D=1$$

$$(A_2 = R) \text{ AND } (A_4 = M) \Rightarrow (D=2)$$

$$((A_2 = R) \text{ AND } (A_1 = M)) \text{ OR } (A_2 = R) \text{ AND } (A_4 = M) \Rightarrow (D=2)$$

$$(A_4 = L) \Rightarrow (D = 3)$$

$$(A_1 = H) \text{ OR } (A_4 = L) \Rightarrow (D=3)$$

$$(A_1 = H) \text{ OR } (A_2 = N) \text{ OR } (A_4 = L) \Rightarrow (D=3)$$

A ₁	A ₂	A ₄	D
M	L	M	1
M	L	H	1
L	L	M	1
L	R	M	2
M	R	M	2
L	R	L	3
H	R	L	3
H	N	L	3

?

Decision Rules

- Final Ruleset:

$(A_2 = L) \Rightarrow (D = 1)$

$(A_4 = H) \Rightarrow D=1$

$(A_2 = R) \text{ AND } (A_4 = M) \Rightarrow (D=2)$

$(A_2 = R) \text{ AND } (A_1 = M) \Rightarrow (D=2)$

$(A_4 = L) \Rightarrow (D = 3)$

$(A_1 = H) \Rightarrow (D=3)$

$(A_2 = N) \Rightarrow (D=3)$

Here, Training Accuracy = 100%

A ₁	A ₂	A ₄	D
M	L	M	1
M	L	H	1
L	L	M	1
L	R	M	2
M	R	M	2
L	R	L	3
H	R	L	3
H	N	L	3

?

Decision Rules

Feature selection algo-1 based Rules

$(A_2 = L) \Rightarrow (D = 1)$

$(A_4 = H) \Rightarrow D=1$

$(A_2 = R) \text{ AND } (A_4 = M) \Rightarrow (D=2)$

$(A_2 = R) \text{ AND } (A_1 = M) \Rightarrow (D=2)$

$(A_4 = L) \Rightarrow (D = 3)$

$(A_1 = H) \Rightarrow (D=3)$

$(A_2 = N) \Rightarrow (D=3)$

Feature selection algo-1 based Rules

$(A_2 = L) \Rightarrow (D = 1)$

$(A_4 = H) \Rightarrow (D=1)$

$(A_2 = R) \text{ AND } (A_4 = M) \Rightarrow (D = 2)$

$(A_4 = L) \Rightarrow (D = 3)$

$(A_2 = N) \Rightarrow (D=3)$

Original Dataset =>

A ₁	A ₂	A ₃	A ₄	D
M?	L?	3?	M?	1?
M?	L?	1?	H?	1?
L?	L?	1?	M?	1?
L?	R?	3?	M?	2?
M?	R?	2?	M?	2?
L?	R?	3?	L?	3?
H?	R?	3?	L?	3?
H?	N?	3?	L?	3?

?

Here, Training Accuracy = 100% for both the classifiers